MILITARY SPECIFICATION

INITIATOR, RF FILTERED, DUAL-BRIDGEWIRE

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of RF filtered, dual-bridgewire initiator for a rocket motor igniter, herein referred to as the initiator.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following standards of the issue listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-220	Method of Insertion-Loss Measurement
MIL-STD-453	Inspection, Radiographic

2.1.2 Other Government documents, drawings, and publications. The following drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

1131AS210	Initiator, RF Filtered

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PUBLICATIONS

Naval Sea Systems Command (Code Ident 53711)

NAVORD Report 2101	Statistical Methods Appropriate for Evaluation of Fuze Explosive-Train Safety and Reliability

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from or as directed by the contracting officer).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Code of Federal Regulations

49 CFR 100-199	Transportation

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington D.C. 20402.)

2.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The initiator assembly shall conform to Drawing 1131AS210 and the requirements herein.

3.2 Radiographic. The radiographic requirements shall conform to the following:

a. The crimping or forming of sections shall be of smooth transition without cracks or unintended deformation.
b. The presence of pyrotechnic charges, spacers, charge holders, and other internal parts shall be verified. The bridgewires shall be in contact with the initiation charge.
c. There shall be no inclusion of foreign matter or evidence thereof.

3.3 Bridgewire circuit resistance. The circuit resistance of each bridgewire shall be 1.0 +0.2, −0.1 ohm. The two bridgewires shall have a common termination.

3.4 Hermetic seal leakage. The leakage rate of the hermetic seal shall be not greater than 1×10^{-6} cubic centimeters (cm^3) of helium (standard temperature and pressure) per second at 1 atmosphere differential pressure.

3.5 Insulation resistance. The insulation resistance shall be not less than 100 megohms at an impressed voltage of 500 volts direct current (Vdc) measured prior to the installation of the RF filter.

3.6 Static discharge sensitivity. The initiator shall be insensitive to static discharge and shall not fire when connected in series with a 5000 ohm resistor and a 500 picofarad capacitor charged to 25,000 volts.

3.7 No-fire current and power. The maximum no-fire current (see 6.3) and power shall be a minimum of 1 ampere or 1 watt per bridge circuit applied simultaneously.

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3.8 Peak pressure output. The peak pressure output in a 10 cm^3 bomb shall be not greater than 1550 nor less than 1050 pounds per square inch gage (psig) as determined in 4.4.8.

3.9 Time to first pressure. The first indication of pressure shall be not greater than 7 milliseconds when functioned with a firing current of 4.5 amperes per bridgewire circuit.

3.10 RF filter effectiveness. The RF filter shall provide a terminated power loss in decibels (dB) at least as effective as that shown in table I.

TABLE I. Terminated power loss in dB.

	Frequency in MHz				Frequency in GHz				
	1	2.5	5	10	27	1.5	2.7	5.6	8.9
Pin-to-pin (1 ohm)	5	13	20	30	55	50	50	50	50
Pin-to-case (0.5 ohm)	9	18	30	40	60	50	50	45	56

3.11 Insertion loss. The RF filter shall provide an insertion loss in decibels with 50 ohm termination at least as effective as that of table II.

TABLE II. Insertion loss in dB.

	Frequency	
	1 MHz	1 GHz
Pin-to-case (50 ohms)	44	70

3.12 Filter capacitance. The capacitance of each filter element shall be less than 0.5 microfarad.

3.13 Marking. The markings shall be legible and shall conform to Drawing 1131AS210.

3.14 Workmanship. The initiator shall be free of blemishes, tool marks, sharp edges, or burrs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Inspection conditions. Unless otherwise specified in the contract (see 6.2.1), inspection conditions shall be as specified in the applicable test method of 4.4.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in 4.4. Failure of any sample to conform to the functioning tests shall be cause for rejection of the lot.

4.3.1 Lot. A lot shall consist of initiators fabricated with inert components produced from the same drawings and specifications in essentially a continuous production run and of pyrotechnic materials blended or mixed in a single batch or lot.

4.3.2 Sampling.

4.3.2.1 Nondestructive examination. The following nondestructive examinations shall be 100 percent (except where noted otherwise).

a. Visual acceptance criteria
b. Radiographic acceptance criteria
c. Hermetic sealing leakage
d. Insulation resistance
e. Bridge circuit resistance
f. Static discharge sensitivity
g. Insertion loss, one percent sample
h. Filter capacitance

4.3.2.2 Functioning tests. Sampling for functioning tests shall be in accordance with the following:

a. Non-fire and power: Forty-five initiators shall be used to provide and estimate of the maximum no-fir current and power with a reliability of 99 percent at a confidence level of 95 percent, based on NAVORD Report 2101.
b. Samples used in establishing the maximum no-fire current and power that are not fired shall be furnished to the procuring activity and shall be so identified. Furthermore, they shall not be considered part of any deliverable lot of acceptable initiators.
c. Peak pressure output: Sampling for peak pressure output testing shall be in accordance with table III.

TABLE III. Sampling for pressure output.

Lot size	Number of samples
Up to 800	10
801 and greater	25

4.4 Test methods.

4.4.1 Visual inspection. Workmanship shall be determined by visually inspecting the initiator assembly.

4.4.1.1 Dimensional inspection. All dimensions shall conform to Drawing 1131AS210.

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4.4.2 Radiographic inspection. Radiographic inspection shall be determined in accordance with quality level 2 of MIL-STD-453 under the following conditions:

a. Inspect each initiator with a maximum of one straight through radial exposure perpendicular to the longitudinal axis.
b. Identify each initiator with its serial number. The part and lot number shall appear on each plate.

4.4.3 Bridgewire circuit resistance. Each bridgewire circuit resistance shall be determined using a maximum test current of 10 milliamperes.

4.4.4 Hermetic seal leakage. Hermetic seal leakage shall be determined in accordance with the following:

a. Subject the initiator to a maximum pressure of 25 milliamperes of mercury for a minimum of 5 minutes, then to helium at a minimum pressure of 30 psia for a minimum of 30 minutes.
b. Determine the leakage from the cartridge when measured at a minimum of 1 atmosphere differential pressure within 20 minutes after removal from the helium environment.

4.4.5 Insulation resistance. Insulation resistance shall be determined prior to installation of the RF filter in accordance with MIL-STD-202, method 302, test condition B, and test shorted pins to body using 15 seconds minimum electrification time. NOTE: This test shall be performed prior to the installation of the RF filter.

4.4.6 Static discharge sensitivity. Static discharge sinsitivity shall be determined by discharging a 500 ± 5 percent picofarad capacitor charged to 25,000 ± 500 volts and 5,000 ± 5 percent ohm resistor shall be connected in a 5 microhenry total inductance series circuit between pairs of pins or leads in all combinations and between the shorted pins or leads (all pins or leads shorted toeach other external to the initiator) and the case of the initiator. The series connection shall be maintained for 60 seconds. Switching in this circuit shall be accomplished by bringing together two 0.5 ± 0.05 inch spherical metal electrodes from and initial separation of 3 inches in air. Each series test shall constitute a separate test. Initiators used for this test shall be temperature conditioned for a minimum of 12 hours at 70° ± 5°F.

4.4.7 No-fire current and power. No-fire current and power shall be determined in accordance with the following:

a. Perform the tests at 75° ± 5°F.
b. Starting at a preselected current level (nominally 2.0 ampere/bridgewire), apply current simultaneously to both bridgewires of one initiator for a minimum of 5 minutes or until the initiator fires. For each succeeding unit, incrementally decrease the current (if the preceding initiator fired) or increase the current (if the preceding initiator did not fire). The change in current shall be in increments of 0.1 ampere. Only one current level shall be allowed for each initiator tested.
c. Statistically analyze and determine the maximum no-fire current in accordance with NAVORD Report No. 2101, Bruceton Method of Analysis.
d. Compute the maximum no-fire power using the following formula:

$$P_{max} = 0.9\ (I_{max})^2$$

where: P_{max} is the maximum no-fire power
I_{max} is the maximum no-fire current

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4.4.8 Peak pressure output and time to first pressure. Peak pressure output and time to first indication of pressure shall be determined in accordance with the following:

a. Function the initiator in a 10 cm^3 bomb at 75° ± 15°F with a current of 4.45 to 4.50 amperes per bridgewire.
b. Monitor pressure time and record firing current, peak pressure, zero time to fisrt indication of pressure, and zero time to peak pressure.

(1) Determine the mean peak pressure output and standard deviation of the sample.
(2) For a sample size of 10, apply = 2.821 standard deviations to the sample mean peak pressure output, based on NAVORD Report 2101 and the T-value for 99 percent confidence level.
(3) For a sample size of 25, apply = 2.492 standard deviations to the sample mean peak pressure output, based on NAVORD Report 2101 and the T-value for 99 percent confidence level.
(4) Compare the upper and lower values obtained in 2 or 3 with the requirements of 3.8.
(5) Compare the time to first indication of pressure with the requirement of 3.9.

c. Record the above results, the individual values, sample mean peak pressure output, and standard deviation.
d. Record the individual results for time to first indication of pressure.

4.4.9 Insertion loss. Insertion loss shall be determined in accordance with MIL-STD-220.

4.4.10 Filter capacitance. Prior to installation of the bridgewires, the capacitance of each filter element shall be determined between each pin and the case of the initiator.

4.4.11 Packaging and marking inspection. Packaging and marking of the initiator shall conform to the requirements of Drawing 1131AS210.

5. PACKAGING

5.1 Packaging and packing. Packaging and packing shall be level C and in accordance with 49 CFR 170-178.

5.1.1 Level C. The initiators shall be packaged and packed in containers complying with the rules and regulations applicable to the mode of transportation. As a minimum, protection shall be such as to prevent deterioration of the material during shipment and ensure safe delivery at destination.

5.2 Marking.

5.2.1 Special marking. The external packings shall be marked with labels for Class C explosives in accordance with 49 CFR 170-178.

5.2.2 Normal marking. In addition to any special marking specified in the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The RF filtered initiator is intended to be used as the initiating device for a rocket motor igniter where high RF energy is present.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Inspection conditions, if other than as specified in 4.2
c. Request delivery or unfired initiators used in no-fire tests (4.3.2 b).
d. Special packing or marking, if required (see 5.1 and 5.2).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52-209-3 and -4 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4.8	Test Reports	DI-T-2072	

(Data item descriptions related to this specification, and identified in this section will be approved and listed as such in DoD 5000.19L., Vol.II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 Definitions.

6.3.1 Maximum no-fire current. Maximum no-fire current is the greatest stimulus that does not cause initiation within 5 minutes of more than one percent of the initiators at a confidence level of 95 percent.

Preparing Activity

NAVY - AS
(Project 1377-N758)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-85627(AS)	INITIATOR, RF FILTERED, DUAL-BRIDGEWIRE

3a. NAME OF SUBMITTING ORGANIZATION

Naval Ordnance Station
Airborne Weapons Engineering Division

b. ADDRESS *(Street, City, State, ZIP Code)*

Indian Head, MD 20640

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☒ OTHER *(Specify)*: Navy

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

NEW SPECIFICATION BEING SUBMITTED.

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
Brown, Fred Code 5212I	(301) 743-4331
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	**8. DATE OF SUBMISSION** *(YYMMDD)*
Naval Ordnance Station Indian Head, MD 20640	84JL19

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